28 March 1974
SUPERSEDING
MIL-B-45529A(WC)
26 September 1966

MILITARY SPECIFICATION

BIPOD, RIFLE: M2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of bipod designed for use with the 7.62mm, M14 type rifles to support the muzzle end of the rifle during firing.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification For.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.
MIL-P-14232	-	Parts, Equipment and Tools For Army Material, Packaging and Packing of.

STANDARDS

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.

FSC 1005

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DRAWINGS

U.S. Army Weapons Command

C7790769	-	Combination Tool.
D7790688	-	Bipod, Rifle: M2.
D11010038	-	Sling, Small Arms
IEL7274621	-	Index of Inspection Equipment Lists.

PUBLICATIONS

U.S. Army Weapons Command

P7790688	-	Packaging Data Sheet for Bipod, Rifle: M2.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

* 3.2 Materials, construction and design. Bipods shall conform to the materials, construction, and design requirements specified herein, on Drawing D7790688 and MIL-W-13855.

3.2.1 Jaws and swivel. The left hand and right hand jaws and the swivel shall be retained in the head assembly by the headed pin which in turn shall be retained by the cotter pin. The jaws shall move to the stop positions and the swivel shall rotate without binding.

3.2.2 Leg assemblies. The leg assemblies shall be capable of being pivoted manually between the open and folded positions without binding on the yoke assembly when the pivot plungers are depressed.

3.2.3 Leg extension assemblies. The leg extension assemblies shall be capable of being moved between the closed and extended positions without binding on the leg shaft assemblies when the leg extension plungers are depressed.

3.2.4 Plungers. The pivot plungers and the leg extension plungers shall move between the extended and compressed positions under spring

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action without binding, and when indexed and released shall be returned to and held in the extended position by spring action. The pivot plungers shall lock the leg assemblies in the open or folded positions and the leg extension plungers shall lock the leg extension assemblies to the leg shaft assemblies in any of the five locking positions. The leg assemblies shall remain in the positions set until the plungers are manually depressed and the leg assemblies are manually reset.

3.3 Performance.

* 3.3.1 Functioning. Bipods shall be capable of being securely attached to, and readily detached from an M14 rifle. When the bipod is assembled to the rifle the jaws shall rotate to the stop positions and the legs shall be capable of being fully opened, folded, and telescoped without binding on the rifle. The rifle sling (Drawing No. D11010038) shall be capable of being assembled to and securely retained by the swivel. Testing shall be as specified in 4.5.3.2.

* 3.3.2 Endurance. While attached to an M14 rifle, bipods shall be capable of withstanding the firing of 3,000 rounds (2,000 rounds with the bipod legs fully opened and extended and 1,000 rounds with the legs opened and fully closed) of Government standard 7.62mm, M80 cartridges without parts deformation, unserviceable parts, or cracking of welded or brazed joints. The rifle shall be set for automatic firing and fired in bursts of approximately 5 rounds each. Retightening of the bipod locking screw shall be allowed after the first and second 500 rounds of firing. After completion of endurance firing, no more than three-fourths of a turn of the bipod locking screw shall be required to retighten the bipod to the rifle. Testing shall be as specified in 4.5.3.3.

* 3.4 Interchangeability. Unless otherwise specified on the drawings, all replacement parts shall be interchangeable. (In normal assembly operations, there shall be no objections interposed to preferential assembly of parts provided all parts are dimensionally and functionally acceptable.) Testing shall be as specified in 4.5.3.4.

3.5 Marking. Bipods shall be marked in accordance with MIL-W-13855.

3.6 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s))shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Bipods shall be assembled from component parts that have met all inspection requirements specified herein. Endurance test lot size shall be as specified in 4.5.3.3.1.

4.5.2 Examination.

* 4.5.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be as specified in the contract (see 6.1). The contractor's examination of these parts shall be prior to assembly into the end item or acceptance as repair parts.

* 4.5.2.2 Bipods. Each bipod in the inspection lot shall be examined in accordance with Table I and the following provisions:

a. Where "Visual" is specified as the inspection method for assemblies, the assembly shall be visually examined for conformance to the requirements specified on the applicable drawings.
b. Where "Visual" is specified as the inspection method for functioning requirements, the assembly shall be visually examined for completeness and manually operated for functioning requirements.

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c. The examination provisions should be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Re-inspections of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained.

TABLE I

	Reference Paragraph	Inspection Method
Materials, construction and design (MCD)	3.2	Visual
Jaws and Swivel	3.2.1	Visual
Leg Assemblies	3.2.2	Visual
Leg Extension Assemblies	3.2.3	Visual
Plungers	3.2.4	Visual

4.5.3 Testing.

* 4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete bipod. If test requirements cited herein are not met, acceptance of the bipod shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional physical and visual examination of the components which are suspected to be the cause of failure or malfunction.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine bipods, partially assembled bipods, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned bipods for retest.

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* 4.5.3.2 Functioning testing. Samples of bipods shall be taken from each inspection lot in accordance with MIL-STD-105 and tested using the test method specified in 4.6.1 for functioning (see 3.3.1). Acceptance shall be based on an AQL of 0.65 percent defective. Failure of the functioning test shall be cause for rejection of the inspection lot and the contractor shall perform corrective action as specified in 4.5.3.1.

* 4.5.3.3 Endurance testing. Two bipods shall be selected at random from the first half of the total production quantity and one of the two shall be tested for endurance (see 3.3.2) using the test method specified in 4.6.2. Failure of the bipod to meet the requirements shall cause a retest of the remaining bipod. If the second bipod fails to meet the requirements the represented lot shall be rejected and the contractor shall perform the corrective action as specified in 4.5.3.1. The same procedure shall apply to the second half of the production quantity.

4.5.3.4 Interchangeability testing.

* 4.5.3.4.1 In plant.

4.5.3.4.1.1 Bipods. A sample of ten bipods selected at random from each inspection lot shall be tested for interchangeability (see 3.4) using the test method specified in 4.6.3.1.1. Bipods taken for interchangeability testing shall have been found satisfactory in all other examinations and tests except endurance testing. Test frequency may be reduced to not less than one test of ten bipods each month when a record of consistently satisfactory results has been established. The ten bipods shall be tested for and shall comply with the requirements for functioning after interchange of parts using the test method specified in 4.6.1. Hand refinement of parts will be allowed on not more than two bipods during interchanging of parts provided that no part is altered beyond drawing requirements. Failure of bipods to meet the requirements of the interchangeability test or failure of any bipod to meet the requirements of the functioning test shall be cause for retest or rejection of the represented lot. At the discretion of the Government representative, an interchangeability retest may be allowed without reconditioning the lot of bipods. Failure in the retest shall cause rejection of the represented lot and the contractor shall perform the corrective action specified in 4.5.3.1. A sample of 20 bipods from each retest or reconditioned lot shall be tested using the same procedure described above, except that hand refinement of parts will be allowed on not more than four bipods.

* 4.5.3.4.1.2 Concurrent repair parts. At least two parts from each inspection lot shall be subjected to the test method specified in 4.6.3.1.2. Failure of any part to meet the requirements shall be cause for rejection of the represented lot and the contractor shall perform the corrective action as specified in 4.5.3.1. Parts subject to reconditioning and further test shall become a reconditioned lot and a sample double the

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number of parts used in the original test shall be tested from each reconditioned lot using the test method specified in 4.6.3.1.2.

4.5.3.4.2 Interplant. When bipods are manufactured concurrently by more than one contractor, each contractor shall forward monthly six bipods for the interplant interchangeability test specified in 4.6.3.2. (see 6.1). The contractor will be informed of the results of the test which indicates failure of the bipods or parts to meet prescribed requirements.

4.5.3.5 Component parts and concurrent repair parts testing. Raw material testing, part testing, and certification shall be as specified in the contract (see 6.1). This will include chemical analysis and physical tests of materials, and tests of protective finishes, heat treatment, bonding, and function of parts as applicable. These tests shall be accomplished prior to assembly of parts into the end item.

* 4.5.3.6 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to applicable drawings, specification of the materials and processes specified on Drawing D7790688.

* 4.5.3.7 Packaging examination and testing. Unless otherwise specified (see 6.1), the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.4 Inspection equipment.

* 4.5.4.1 Acquisition, calibration, maintenance and disposition. Unless otherwise specified (see 6.1), responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment prescribed on the List of Inspection Equipment, IEL7274621, and for all other inspection equipment required by applicable specifications, shall be in accordance with MIL-I-45607.

* 4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test Methods.

* 4.6.1 Functioning test. Bipods shall be tested using an M14 type rifle or a Government approved, contractor designed, test fixture. The rifle or fixture shall have maximum material conditions of mating rifle surfaces. The bipods shall be assembled to the rifle or fixture by manually tightening the locking screw using the combination tool conforming to Drawing C7790769. The bipod shall then be checked for secure

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retention on the rifle or fixture, rotation of the jaws to the stop positions when the legs are in the open position, and to insure that the legs are capable of being opened, folded, and telescoped without binding. The rifle sling (Drawing D11010038) shall be assembled to the bipod swivel and checked for retention. Disassembly of the bipod from the rifle or fixture shall be accomplished by loosening the locking screw to unlock the jaws. After disassembly, the bipod jaws shall be magnetic particle inspected for cracks, seams or other injurious defects in accordance with the applicable drawings.

* 4.6.2 Endurance test. Bipods shall be securely attached to an M14 type rifle using the combination tool conforming to Drawing C7790769 (see 6.1). The bipod legs shall be locked in the open position and shall be firmly in contact with packed dirt during firing of the rifle. Firing shall be done from a prone or standing position or from a Government approved fixture of the contractor's design. The rifle shall be fired 2,000 rounds with the bipod legs fully extended and 1,000 rounds with the bipod legs fully closed. The rifle shall be set for automatic firing (the selector set with the letter A to the rear of the rifle) and fired in bursts of approximately 5 rounds each. Retightening of the bipod lock screw shall be allowed after the first and second 500 rounds of firing. The bipod shall be visually and manually checked for: secure retention on the rifle, (no more than three-fourths of a turn required to retighten the bipod to the rifle) deforming of parts, unserviceable parts, and cracking of brazed joints. Welded joints and the jaws shall be checked for cracks using the magnetic particle inspection method specified on the applicable drawings. At the completion of the endurance test, bipods shall be disposed of as specified in the procurement documents (6.1).

4.6.3 Interchangeability Test.

4.6.3.1 In plant.

* 4.6.3.1.1 Bipods. Bipods shall be tested for interchange of parts by disassembling and then reassembling parts using the parts and prearranged system specified below. Interchange of parts shall be accomplished by dividing the parts of each bipod into 10 groups of nonmating parts as shown below and distributing the groups into 10 different trays until each tray contains parts for a complete bipod. Groups of parts from the first bipod shall be taken in order and placed in trays 1 through 10; groups of parts from the second bipod shall be taken in order and placed in trays 2 through 10 to 1; groups of parts from the third bipod shall be taken in order and placed in trays 3 through 10 to 2; etc. Commercial parts such as screws, spring pins, etc., shall be placed in the same tray as their mating or associate part. Any commercial

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part rendered unserviceable by disassembly shall be replaced without penalty to the interchangeability test. Any noncommercial part rendered unserviceable by disassembly or assembly shall be cause for failure of the test. After interchange of parts, each reassembled drum shall be function tested in accordance with 4.6.1.

GROUPS OF NON-MATING PARTS

GROUP I

Button, plunger, pivot (7790820) (2 required)

GROUP II

Jaw, L H (7791102)

GROUP III

Leg assembly, bipod, R H (7790822) (without plunger 7790836 and spring 7790838)

GROUP IV

Pin, straight, headed (7791669) (used with yoke 7791106)
Plunger, pivot (7792846) (2 required)

GROUP V

Spring, helical, compression (7790824) (2 required) (used with plunger 7792846)

GROUP VI

Head assembly, bipod (7792847)
Spring, helical, compression (7790838) (2 required) (used with leg assemly 7790821 and 7790822)

GROUP VII

Plunger, leg extension (7790836) (2 required)

GROUP VIII

Jaw, R.H and screw assembly (7791107)

GROUP IX

Leg assembly, bipod, L H (7790821) (without plunger 7790836 and spring 7790838)

GROUP X

Swivel, gun sling (7791670)

4.6.3.1.2 Concurrent repair parts. Concurrent repair parts shall be tested by disassembling two or more bipods, previously tested in 4.5.3.4.1.1, and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed, and the bipods shall operate and function properly. This test may be performed independently of the bipod interchangeability test specified in 4.5.3.4.1.1, and at more frequent intervals using accepted bipods taken from current production.

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4.6.3.2 Interplant. Bipods to be subjected to the interplant interchangeability test shall be given a preliminary hand functioning to assure proper operation before parts are interchanged. Parts shall then be disassembled from the bipods, and identified as to manufacturer. Bipods shall be interchanged in a manner similar to the detailed plan specified in 4.6.3.1.1 except that parts shall be divided into six groups and that when disassembling, every other bipod used shall be one produced by a different manufacturer. After interchange of parts, the bipod shall be function tested in accordance with 4.6.1. Before bipods are returned to the contractors, the original parts shall be reassembled to their respective bipods.

5. PREPARATION FOR DELIVERY

* 5.1 Pilot pack. The pilot pack shall consist of a complete and unit packed shipping container in accordance with Packaging Data Sheet P7790688 for the level of protection specified in the contract (see 6.1) and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing, and marking. Bipods shall be preserved, packaged, packed and marked in accordance with the requirements of Packaging Data Sheet P7790688 for the level of protection specified in the contract (see 6.1).

* 5.3 Repair parts. Repair parts shall be prepared for delivery in accordance with applicable packaging data sheets or as specified in the contract or order.

6. NOTES

* 6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Lists of drawings and specifications pertinent to the bipods, showing applicable revision dates.
c. Index of inspection equipment lists pertinent to the bipods, showing applicable revision dates.
d. Shipping instructions for First article (see 3.1 and 5.1).
e. Criteria for inspection of component parts and concurrent repair parts (see 4.5.2.1 and 4.5.3.5).
f. Disposition of endurance tested bipods (see 4.6.2).
g. Packaging examination and testing, if different (see 4.5.3.7).
h. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).

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i. Packaging instructions for repair parts (see 5.3).
j. List of acceptance inspection equipment to be furnished the contractor (see 4.5.4.1) and responsibilities for other Government property to be furnished the contractor.
k. Responsibilities for furnishing ammunition, combination tool, rifle, and sling for the functioning and endurance tests (see 4.6.1 and 4.6.2).
l. Shipping instructions for bipods when an interplant interchangeability test is required (see 4.5.3.4.2).

* 6.1.2 Contract data requirements. Requirements for monthly reports of the results of final examination, functioning, endurance and interchangeability tests shall be specified on a DD Form 1423 included in the contract.

* 6.2 Unless otherwise specified (see 6.1.1 j and 4.5.4.1), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

* 6.3 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

* 6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - WC

Preparing Activity:
Army - WC

Project Number:
1005-A515

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-B-45529B; Bipod, Rifle: M2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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